CHAGEE

CHAGEE SINGAPORE

THE OPPORTUNI-TEA — UNLOCKING GROWTH AT CHAGEE SG WITH AI

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



SPEAKER NOTES · TITLE

Good morning. We are Group 3. Our company is Chagee in Singapore, and our story has three solutions: online retailing, matching each online order with a convenient pickup; in-store management, serving that demand well at the counter; and new store expansion, opening where customers are underserved.

Each solution has its own decision-maker: the digital commerce manager with operations, the operations manager or outlet GM, and the expansion manager. After the close, an appendix holds the detail for questions.

Timing: title and agenda together, about twenty seconds.

AGENDA

Spilling the Tea

01 **Introduction**
Chagee Singapore, our scope and why

02 **Executive Summary and Business Plan**
Two key challenges and one strategic decision

03–08 **Our Business Recommendations**
Dynamic demand management · AI-enabled in-store planning · AI-enabled new-store site selection

09 **Execution Plan**
How can we operationalise this business plan

10 **Business Impact**
Financial and non-financial benefits of the business plan

11 **Q & A**

\+ **Appendix**
Supporting evidence



SPEAKER NOTES · AGENDA

Not presented: go straight from the title to the introduction. For reference: the introduction and the executive summary; our recommendations as three solutions, each with its decision-maker (online retailing for the digital commerce manager with operations, in-store management for the operations manager or outlet GM, new store expansion for the expansion manager); the execution plan, which is our recommendation slide with the pilots and when to proceed; and Q&A. Business impact has no main slide yet: the line opens the draft in the appendix. The Australia extension sits after Q&A, just before the appendix, which now also holds the data trail, the earlier scope slide and the membership chapter. Click any line to jump there.

Today, we're presenting Chagee

A tea brand most of you know: order in the app, collect at the counter, or have it delivered to your location.



Our scope

Chagee Singapore's management and its customers

Outlets are **company-owned, not franchised**, so the counter, roster and equipment are Chagee's own calls.

Why Chagee?

1. **Many stores, managed by one company**, one data system, with lots of data
2. **App-centric**: collects personal and transactional data (their rich data is our oil)
3. **Fast-growing, playful**: rapid expansion, room for improvements with AI
4. **An insider's view**: we were fortunate to interview a staff member (anonymous)

FROM FARM TO CUP · OUR PROJECT FOCUS IS DOWNSTREAM

Source
Yunnan tea; local milk and fruit

Warehouse
Restocking from forecasts

Upstream data and decisions sit with **Chagee corporate**

Outlets
Make and hand over

via Chagee App

Last mile
Collection or delivery

Downstream data is analysed and decisions are made by **Chagee SG** (APAC HQ)

OUR FOCUS | **The downstream operational challenges: some of the most pressing issues for Chagee Singapore's management.**

SPEAKER NOTES · CHAGEE SINGAPORE AND OUR SCOPE

Today we are presenting Chagee: a tea brand most of you know. Order in the app and collect at the counter, or have it delivered; the app is for ordering ahead, and delivery goes through third-party apps such as Grab (anonymous Chagee source).

Our scope is Chagee Singapore, its management and customers. Singapore outlets are company-owned, not franchised, as our project owner confirmed, so the counter, roster and equipment are Chagee's own decisions. If asked: group-wide, about 88% of FY 2025 net revenue came from franchising (secondary source); that is global context, not Singapore.

Why Chagee: many stores under one company and one data system (anonymous Chagee source); app-centric: an order-ahead, self-pickup app (App Store listing), in-app ordering after the reopening queues (Vulcan Post), and personal and transactional data collected (Form 20-F, FY 2025); fast-growing and playful, with room for AI. The app's share of Singapore orders, and whether searches or location are logged, are to confirm. And we interviewed a Chagee staff member, who asked not to be named.

Farm to cup: tea comes from Yunnan through standard channels, only fresh milk and fruit are local (YF Chuhai), and restocking is planned from forecasts (Form 20-F). Downstream, the outlet makes each cup and the app links it to collection or delivery; which of these Chagee records is to confirm. Upstream sits with Chagee corporate; downstream is analysed and decided by Chagee Singapore, so that is our focus. Caution: this split, and the APAC HQ label, come from our team's storyboard with no public source; confirm before presenting.

So we focus on the downstream operational challenges. That leads into the executive summary. About one minute.

Our business plan to solve Chagee's problem statements



CASE STUDY · OUR INTERVIEW

"Mei"

Membership Operations · APAC, Chagee

Persona: name changed, interviewee anonymised

WHAT MEI TOLD US

Through the interview, we learnt about Chagee's **business model**, its **local operations**, and some of its most **pressing concerns and strategic needs**, which AI could potentially help solve.

TWO KEY CHALLENGES + ONE STRATEGIC DECISION POINT

KEY CHALLENGE

Online retailing

Use data for better demand management, to grow sales and manage the customer experience.

PROPOSAL 1

Dynamic demand management

At peak, AI reroutes pickups and deliveries to nearby outlets with room, decreasing drop-offs and wait times, and increasing overall sales.

See how it works →



DECISION-MAKER

Zone and branch manager

KEY CHALLENGE

In-store management

A faster, smoother counter: shorter waits, a crew matched to demand and less wastage.

PROPOSAL 2

AI-enabled in-store planning

Higher-precision outlet and zone forecasts help managers decide on staffing, tea machines and fresh stock to maximise store performance.

See how it works →



DECISION-MAKER

Zone and branch manager

STRATEGIC EXPANSION DECISION

New store openings

Where to open next, so a new store is profitable and adds customers instead of moving them.

PROPOSAL 3

AI-enabled new-store site selection

Picks new sites that add customers and profit.

See how it works →



DECISION-MAKER

Expansion and growth manager

SPEAKER NOTES · EXECUTIVE SUMMARY

Our business plan starts from Chagee's own problem statements. We interviewed a member of Chagee's Membership Operations team for APAC, shown here as a persona, "Mei": the name is changed and the interviewee asked not to be cited openly.

Through the interview, we learnt about Chagee's business model, its local operations, and some of its most pressing concerns and strategic needs, which AI could potentially help solve.

We frame them as two key challenges and one strategic decision point. Challenge one, online retailing: using data for better sales, customer loyalty and cost management, through customer management and reducing wastage. Challenge two, in-store management: a faster, smoother counter, through in-store operations management for shorter waits and a crew matched to demand. And the strategic decision: new store openings, where to open next so a new store is profitable and adds customers rather than moving them from nearby outlets.

The top of each column is what we heard; the shaded part below is our team's proposal, not something Chagee asked for, with its decision-maker in green. Solution one, online retailing: AI-enabled demand management, offering customers an outlet with room and recommending delivery actions for managers to approve, for the digital commerce manager with operations. Solution two, in-store management: AI-enabled in-store planning, forecasting each outlet by day and recommending staff, machines and fresh stock for managers to approve, for the zone manager or outlet GM. The earlier toppings and bucket-brigade exploration is in the appendix. Solution three, new store expansion: identify underserved search clusters, evaluate candidate locations and assess net additional demand and financial outcomes, for the expansion manager. We do not address wastage in this deck; the membership-squad analysis is in the appendix. We walk through them in this order next. About forty-five seconds.

Proposal 1: Dynamic demand management

KEY CHALLENGE 1

Demand management: crowded outlets lose sales, with impacted customer experience



WHAT IT COSTS CHAGEE TODAY

Drop-offs
Customers see the wait and leave without ordering.

Crowding
Those who stay crowd the outlet, so it sells less and delivery orders slow down.

Long waits
Customers wait longer than they expected; the experience suffers.

Delivery orders arrive blind
Delivery customers cannot see the in-store queue, so orders keep coming and riders wait at the counter.

Spoils the in-store image
Chagee's elegant stores are part of the brand. A crowd of frustrated customers at the door undoes the classy experience the brick-and-mortar model is built on.

With AI, this can change

- **Higher sales**
  Demand spread across nearby outlets instead of lost.
- **A better brand image**
  Faster turnaround, well managed, a smart system.
- **Calmer counters**
  Delivery riders do not wait long or crowd the counter.

SPEAKER NOTES · WHEN AN OUTLET IS SWAMPED

Proposal one starts with a problem every Chagee customer has seen: at the peak, one outlet is swamped while another nearby has room. The scene is an illustration, not observed data; treat the points as a problem to investigate with Chagee.

What it costs Chagee today, in five points. One, drop-offs: customers see the wait and leave without ordering, a sale lost before it is placed. Two, crowding: those who stay crowd the outlet, so it sells less and delivery orders slow down. Three, long waits: customers wait longer than they expected. Four, delivery orders arrive blind: delivery customers cannot see the in-store queue, so orders keep coming and riders wait at the counter. Five, it spoils the in-store image: Chagee's elegant stores are part of the brand, and a frustrated crowd at the door undoes that experience.

With AI, this can change: higher sales, as demand is spread across nearby outlets instead of lost; a better brand image; and calmer counters. These are proposed outcomes, not measured Chagee results. About forty-five seconds.

From Chagee's data to demand spread across outlets

An AI agent uses predictive AI to recommend, generative AI explains, and people make the final call.

1 The data

What Chagee collects, or could collect

Chagee app & sales
Orders by outlet and time · Customer location · Margin per drink

Outlet operations
Live queue of open orders · Crew on shift · # of machines

Delivery apps
Delivery orders · Rider waits

Historical events
Decisions made · Profitability outcomes

2 Proposed AI agent

Trigger, recommend, decide, with guardrails

TRIGGER

Wait runs long
A store's wait is, or is about to go, above 40 min.

AI AGENT

Recommend outlet, action, incentive
Predictive AI on live store data and patterns learnt from past decisions, within approved rules.

PEOPLE DECIDE

Customers and managers choose
Generative AI explains each recommendation; they accept or decline.

3 What it does

Example of actions at peak, and it keeps learning

Offers a faster outlet
Routes pickups to the outlet that can hand over sooner.

Recommends a delivery pause
Recommends pausing third-party delivery orders and redirecting demand; the zone manager reviews and approves.

CONTINUOUS LEARNING

Gets smarter over time

Every new decision and its outcome feed back in, so future decisions keep improving.

THE OPERATIONS IDEAS BEHIND IT

Demand shaping	moving demand strategically
Capacity pooling	with nearby outlets
Reducing bottlenecks	dynamically

KEY BENEFITS

- ✓ **Balances the zone**, thereby increasing sales
- ✓ **Learns from past decisions and live data from stores** to raise profit while enhancing customer experience

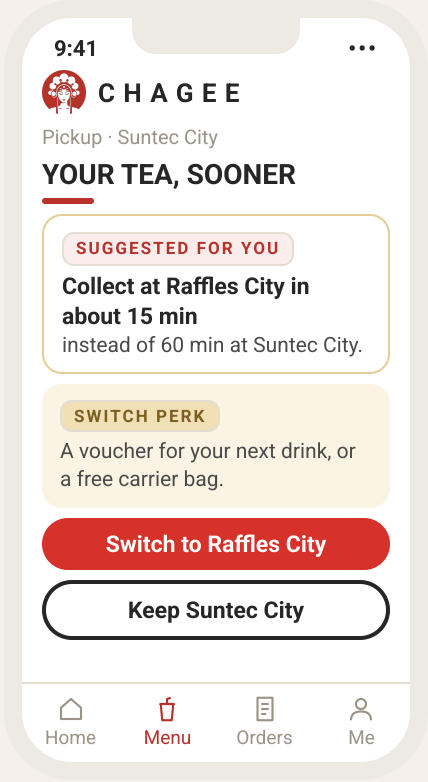
SPEAKER NOTES · PROPOSAL 1: HOW IT WORKS

Section 04, a one-page summary of Proposal 1. The data: Chagee's app and sales, outlet operations, delivery apps and historical events, meaning decisions made and their profitability outcomes. The proposed AI agent: a trigger when a store's wait is, or is about to go, above 40 minutes, an illustrative setting; predictive AI recommends an outlet, an action and an incentive within approved rules; generative AI explains it, and customers and managers accept or decline.

What it does at the peak: offers a faster outlet for pickups, and recommends a delivery pause that the zone manager reviews and approves. Every decision and its outcome feed back in, so the model keeps learning. The operations ideas: demand shaping, capacity pooling with nearby outlets, and reducing bottlenecks dynamically. The benefits we expect, to be tested in the pilot: a better-balanced zone and more sales, and higher profit with a better customer experience. About thirty seconds.

Illustration of Proposal 1: Direct customers to outlets with shorter waiting times.

PICKUP · CHAGEE APP



A faster outlet, offered

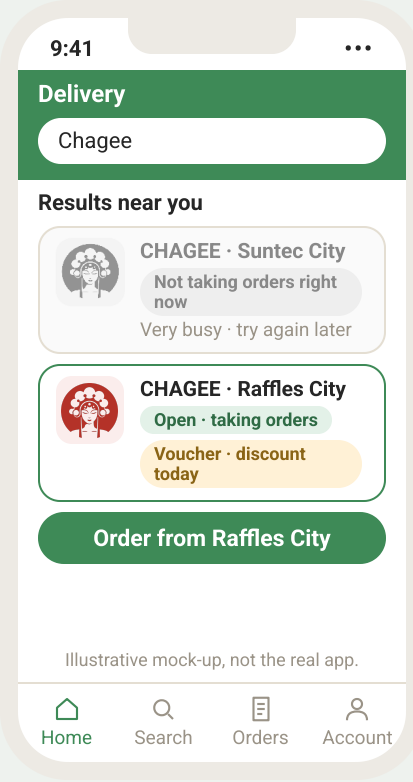
Suntec City is swamped, so the app suggests Raffles City and explains why.

45 min sooner
collect in about 15 min instead of 60, walk included

DYNAMIC PRICING
A voucher or a carrier bag nudges and rewards the switch, offered only while Suntec City is swamped.

Customer chooses before paying.

DELIVERY · E.G. GRABFOOD



The busy outlet steps back

The agent recommends pausing Suntec City on delivery apps and redirecting demand; the zone manager reviews and approves.

Orders flow to Raffles City
no riders stuck at the busy counter

DYNAMIC PRICING
Instead of closing Suntec City on delivery altogether, a voucher at Raffles City can draw demand to the outlet with room.

Zone manager approves the pause and redirection. Placed orders stay put; platform permissions to confirm.

SPEAKER NOTES · THE CUSTOMER EXPERIENCE

What the customer sees, in two channels; switch with the toggle. Pickup, in the Chagee app: the customer near Suntec City sees their outlet and an estimated collection time, a suggested alternative at Raffles City with its collection time, including the walk, and an optional switch incentive. Choose Switch outlet or Keep my outlet: the confirmation and both outlets' workload update. The customer chooses before paying. Tick the fill-up box: once Raffles City is busy too, the saving falls below the approved minimum and the suggestion is withdrawn.

Delivery, in an illustrative third-party app: the agent's recommendation goes to the zone manager first, who can approve a temporary pause at the busy outlet, promote the alternative outlet instead, or decline. Only then does the customer-facing screen change. Orders already placed are not moved, customers are free to order elsewhere or not at all, and the platform, not Chagee, controls its ranking; platform integration and permissions are requirements to confirm.

See illustration opens the pickup race in the appendix: the customer's journey to the faster outlet, step by step; Back to proposal 1 returns here. Every time here is an illustrative scenario from the reference deck, not Chagee data. Then the transition: with demand spread across outlets, how should each outlet plan its staff, machines and fresh stock for the week? About one minute.

Proposal 2: AI-enabled in-store planning to achieve high-performing outlets

KEY CHALLENGE 2

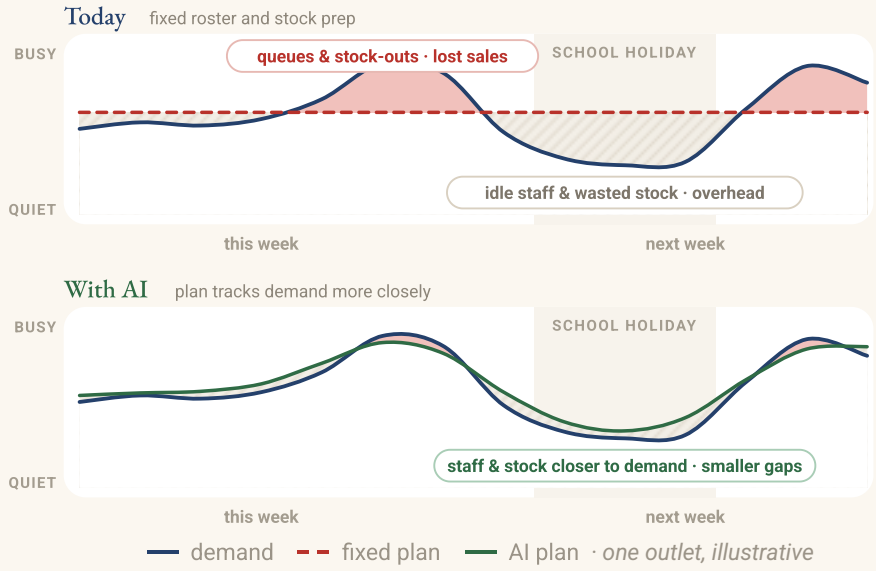
In-store management: matching staff, machines and stock to daily demand

IN PLACE TODAY *Chagee annual report, FY 2025*

- Group-level demand forecasting for stock and inventory
- Centralised, digital store management

ENVISIONED MODEL *enhanced with predictive analytics and AI*

- Forecasts by outlet per day
- AI-recommended rosters, staff moves and machines to run
- Management of fresh inventory, less waste
- Unoptimal store set-ups flagged for review



With AI, this can change

- **Fewer lost sales** — Ready for busy days.
- **Less idle staff** — Rosters flex on quiet days.
- **Less wasted stock** — Fresh prep matches demand.
- **More optimal machinery count** — How many to match demand.

SPEAKER NOTES · AI-ENABLED IN-STORE PLANNING: THE CHALLENGE

Our data and AI enablement initiatives have two parts, and they run on one system. Proposal one is dynamic demand management: it steers customers to capacity in the moment. Proposal two improves the bread and butter of the business, how the store is run: it plans staff, machines and stock ahead of time, while cutting overheads such as idle staff and wasted stock. This is AI-enabled in-store planning to achieve high-performing outlets. The two models are not separate tools: they share one data and AI system and inform each other. Routing decisions from proposal one feed the forecasts in proposal two, and the plans from proposal two tell proposal one where there is room. As more data builds up, every decision gets better. With AI, this can change: fewer lost sales, less idle staff, less wasted stock and a more optimal machinery count.

Click between Today and With AI on the chart. Today, a roster and a daily prep of fresh stock that do not flex day to day: on busy days demand rises above them, so queues form and stock runs out, which loses sales; in the school holiday demand falls below, so staff stand idle and unsold fresh milk and fruit is thrown away. This applies to perishables prepped daily, not tea leaves or packaging that keep. With AI, the plan tracks demand more closely. It will not match demand exactly, but the gaps get smaller: fewer lost sales, less idle staff and less wasted stock.

What is in place today, from Chagee's FY 2025 annual report: group-level demand forecasting for stock and inventory, and centralised digital store management. We build on that. The envisioned model, enhanced with predictive analytics and AI, forecasts each outlet per day, recommends rosters, staff moves and machines to run, manages fresh inventory to cut waste, and flags unoptimal store set-ups for review. Whether Chagee Singapore already does any of these is not confirmed publicly, so we do not claim it doesn't. About one minute.

Managers can use insights to plan and manage stores better

HOW IT WORKS

1. **DATA** — examples: orders · staffing · machines
2. **PREDICTIVE AI** — forecasts each outlet by day
3. **AI PLATFORM** — provides recommendations and a chatbot
4. **MANAGERS** — view insights and plan the week

Chagee Store Insights Dashboard – Zone View · Updated 7am daily

STAFF FOR THE COMING WEEK

OUTLET	RECOMMENDATION
Raffles Place	Reassign 1 crew → City Hall
Suntec City	Maintain current crew
City Hall	Receive 1 crew from Raffles Place

WHY School-holiday pattern learnt from past years: office areas like Raffles Place quieten, leisure areas like City Hall get busier.

UPCOMING PEAKS & QUIET SPELLS

▲ Peak	Suntec City · Fri to Sun, afternoons
▲ Peak	City Hall · weekdays, school holiday
▼ Quiet	Raffles Place · Mon to Fri, school holiday

TEA MACHINES · IN STORE

	CURRENT	RECOMMENDED
Raffles Place	4	3 ▼
Suntec City	4	4 =
City Hall	3	4 ▲

FRESH STOCK · PACKAGES PER WEEK

	CURRENT	RECOMMENDED
Raffles Place	9	7 ▼
Suntec City	8	9 ▲
City Hall	6	8 ▲

One model that keeps getting smarter

1. **Events feed in** — Rerouted pickups, delivery pauses and vouchers from Proposal 1
2. **The model retrains** — on these events and the demand they shifted
3. **Forecasts sharpen** — so this dashboard anticipates better, even after protocol changes

Over time, Chagee manages its stores better, ahead of demand rather than reacting to it.

THE OPERATIONS IDEAS BEHIND IT

- **Demand forecasting** — by outlet and day
- **Capacity planning** — staff, machines and stock to the forecast
- **Throughput** — find and fix the slowest step
- **Staff cross-training** — crew cover different roles / outlets

SPEAKER NOTES · HOW AI-ASSISTED STAFFING WORKS

This is what a zone manager would see each week. It runs on the same data foundation as proposal one: orders, staffing, machines, the calendar, and years of history across all 44 outlets. Predictive AI forecasts each outlet by day; the AI agent turns that into recommendations; managers review and approve.

This is a dashboard the zone manager views, refreshed at seven each morning, here for a City Hall–Marina zone: the One Raffles Place, Suntec City and Raffles City outlets, which we label by area. Take a school-holiday week. For the coming week, it tells the manager whether to reassign staff between outlets, and how many: move one crew member from Raffles Place, where offices quieten, to City Hall, where leisure traffic picks up, and keep Suntec City's crew as it is. Next to it, upcoming peaks and quiet spells to plan around. For the longer term, it compares each outlet's current number of machines with the recommended number: one fewer at Raffles Place, one more at City Hall. And for fresh stock, it compares this week's packages with the recommended weekly amount. The outlets are real; every figure is illustrative, not Chagee data.

On the right, how this connects to proposal one. The two run on one model. Every rerouted pickup, delivery pause and voucher from proposal one is an event the model learns from, together with the demand it shifted. As it retrains, its forecasts sharpen, so this dashboard keeps anticipating demand well, even after protocols change, and plans staff, machines and stock in advance. Over time Chagee manages its stores ahead of demand rather than reacting to it. Behind it are four operations ideas: demand forecasting, capacity planning, throughput, and staff cross-training, so crew can cover different roles when they move between outlets.

Proposal 3: AI-enabled new-store site selection

BUSINESS PROBLEM A new outlet may look successful while mostly moving sales from nearby stores, while other customers have no convenient outlet. **Context** Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). **Simulated** searches below.

1 WHO

Where should we open next?



Which location brings us new customers?
HEAD OF EXPANSION

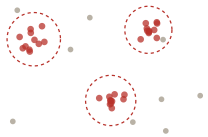
- **A** **Beside an existing outlet**: its sales may come from the store next door.
- **B** **Near underserved customers**: people searching with no outlet nearby.

The manager needs to compare locations, customer access and potential sales—not rely on intuition alone.

2 WHAT

Bring the signals together

- **App searches and unmet demand** — IN THIS DEMO · SYNTHETIC
- **Existing outlets and historical orders** — IN THIS DEMO
- **Residents, workers and transit access** — PROPOSED · NEEDS DATA ACCESS
- **Footfall and nearby competitors** — PROPOSED · NEEDS DATA ACCESS
- **Area delivery demand, if available** — PROPOSED · NEEDS DATA ACCESS
- **Rent and operating costs** — ASSUMED VALUES · P&L VIEW



DBSCAN: unsupervised machine learning that finds dense groups of searches.

These clusters suggest locations to investigate. A separate demand model estimates sales and how much would shift from nearby outlets.

3 RESULT

Compare the options. Make the call.

This site looks promising. Let's compare the assumptions and downside before we commit.
HEAD OF EXPANSION



Site comparison synthetic app data · our assumptions

NET ADDITIONAL CUPS A DAY FOR CHAGEE

B **+731** cluster near Choa Chu Kang

A **+216** beside 34 · West Mall

SALES MOVED FROM NEARBY OUTLETS

B 106 of the 837 cups it sells · **A** 64% of its 604

DEMAND, DELIVERY AND COMPETITION hand-set scores, illustrative pin (appendix)

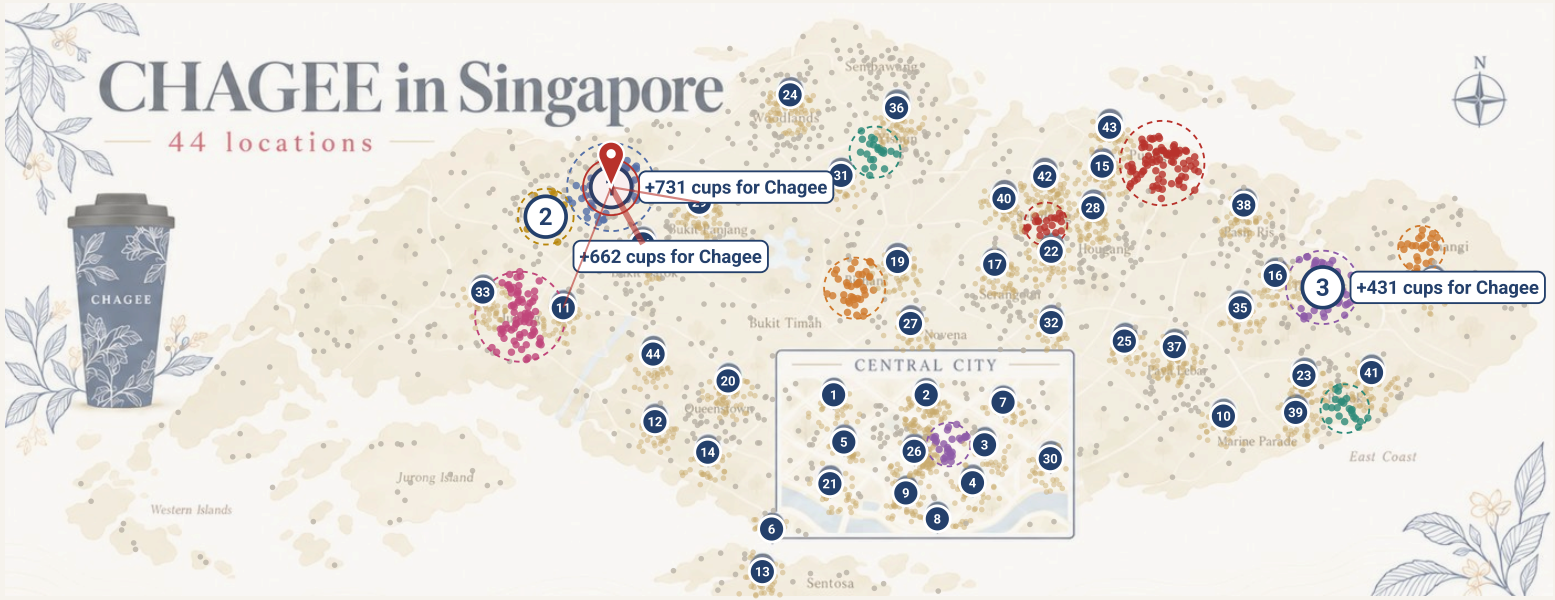
Demand	Delivery	Competition

An AI-assisted dashboard helps the team make an evidence-informed decision. The manager makes the final call.

Proposal 3: AI-enabled new-store site selection

BUSINESS PROBLEM A new outlet may look successful while mostly moving sales from nearby stores, while other customers have no convenient outlet. **Context** Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). **Simulated** searches below.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (counter pages, appendix).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method. Its five-year P&L: appendix.

TAKEAWAY | Our model picks the site that adds the most cups to Chagee: the cluster near Choa Chu Kang, about **+731** a day. Judged on sales alone, the same store beside an existing outlet looks busy (about 604 cups a day) but adds only **+216**: the rest moves from the outlet next door.

SPEAKER NOTES · PREDICT DEMAND FOR A NEW STORE

Solution three, new store expansion. The business problem: Chagee needs locations with enough additional demand and attractive economics. It has two sides: a new outlet can look successful while mostly moving sales from the stores around it, and customers elsewhere may have no convenient outlet at all. Context, from a Chagee source who asked not to be named: Chagee is looking to expand rapidly in Singapore, and how many stores it is going for is confidential. Chagee Singapore runs on its app, with few kiosks. If Chagee logs searches and their location, which we have to confirm, every search is a demand signal, including searches where no store is close. Here the searches are synthetic. That makes the app the forecasting input for a new store, before it has any sales.

Step one: synthetic app sessions on our map. Gold are within walking distance of an outlet; red found no store nearby. Step two: DBSCAN, unsupervised machine learning, groups red searches that are dense; scattered ones are noise. Step three: cluster centres are candidate sites.

Step four: click the map to test a site. The card leads with the gain for Chagee, which is the store's sales minus the cups it takes from nearby outlets, and gives a verdict: strong, mixed, or mostly moving cups. Try a pin next to Tampines West CC: it sells, but most of it moves from the two Tampines outlets nearby. Step five: the forecast's peak hour gives the opening roster for the counter pages in the appendix.

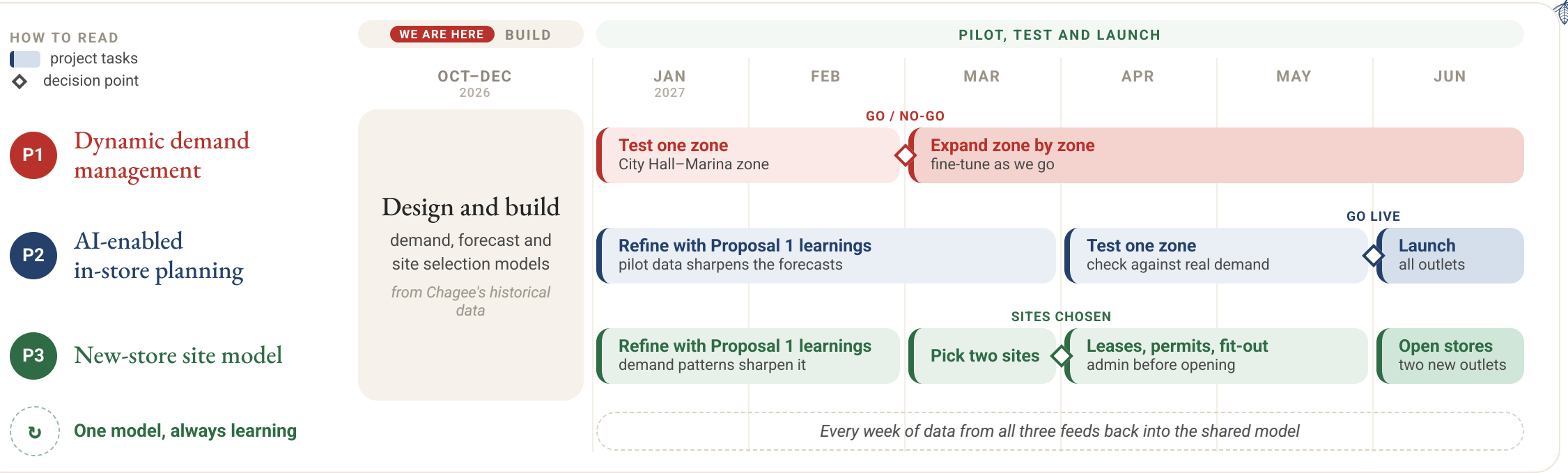
If asked how DBSCAN works, press How it works: it replays the algorithm one search at a time on a toy set. The appendix lists every simulated number.

The three sliders are the only inputs: cluster radius, minimum searches per cluster, and how far people will walk. The data is synthetic; with Chagee's real app logs, the same steps would run unchanged. About one and a half minutes.

EXECUTION PLAN

Three months to plan, then a phased six-month launch

One integrated AI solution: plan from now to December, start small in January, learn, then scale.



PROJECT TEAM — Our team: Project manager, AI engineer + From Chagee: Supply chain manager (APAC HQ), Zone manager, Store manager

By end June 2027

- ✓ **One integrated platform**, with Proposals 1, 2 and 3 launched across Singapore, improving demand management, decision making and site selection
- ✓ **Two new stores** opened in strong Singapore locations, chosen to maximise profit

SPEAKER NOTES · EXECUTION PLAN

Here is how we would phase it as one integrated AI solution. From now to December the project team plans: we design and build the three models: the demand model for proposal one, the forecast model for proposal two and the site selection model for proposal three. Then, from January, six months to launch.

Proposal one, dynamic demand management, starts first. We test it in one zone, the City Hall–Marina zone from the dashboard: Raffles Place, Suntec City, City Hall and Clarke Quay, for two months, then take a go or no-go decision and expand zone by zone, fine-tuning as we go.

Proposal two, AI-enabled in-store planning, is designed and built from now to December. From January, the findings from the proposal one pilot feed in and sharpen the forecasts; we then test them in one zone in April and May and launch in June.

Proposal three, the site selection model, is also designed and built from now to December, on outlet and site history. In January and February, what the proposal one pilot learns about demand refines it, and in March it picks the next two sites, which takes little time once the model is built. April and May are the administration before a store can open, leases, permits and fit-out, and the two new stores launch in June.

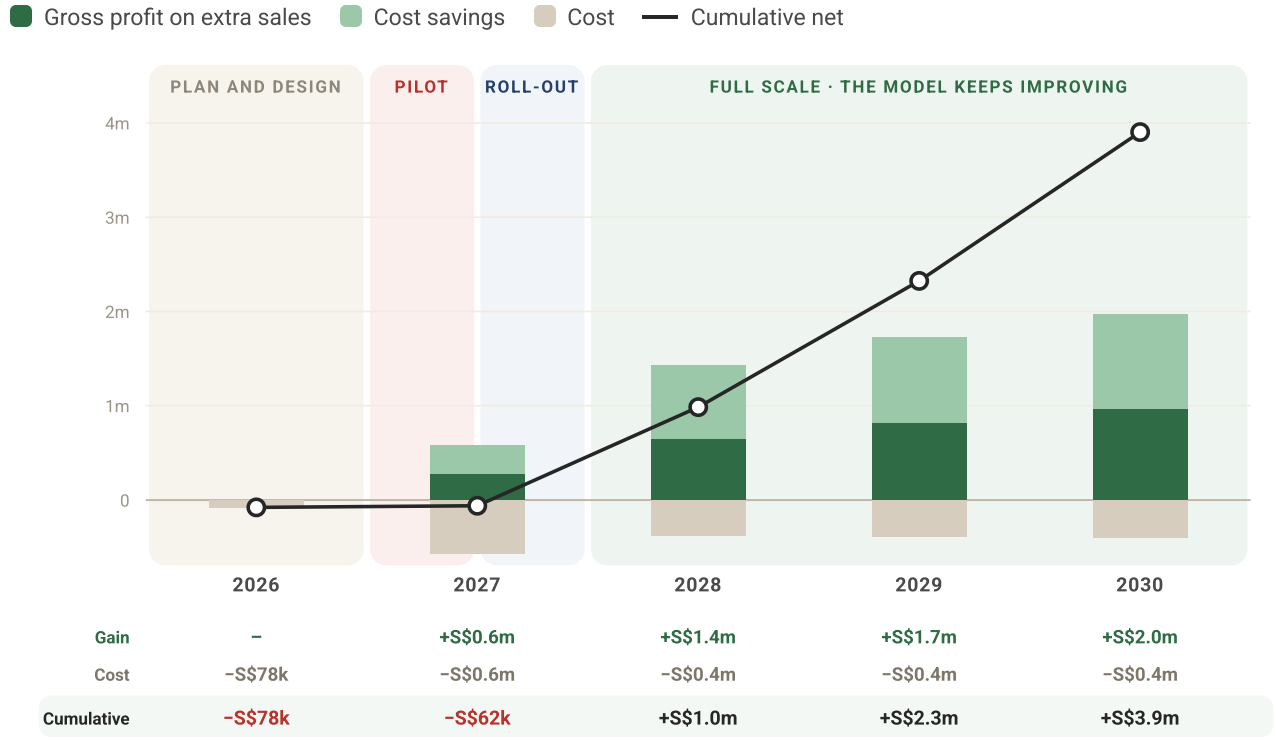
The work is run by a small joint team: our project manager and AI engineer, with Chagee's supply chain manager from APAC HQ, a zone manager and a store manager. Underneath, one model keeps learning from every week of data. By the end of June, Chagee has one integrated platform, with all three proposals launched across Singapore, improving demand management, decision making and site selection, and two new stores opened in strong locations chosen to maximise profit. The twelve-month view is in the appendix. About fifty seconds.

FINANCIAL IMPACT · SINGAPORE · FIVE YEARS

We forecast our business plan to add S$3.9m net profit over five years and break even in early 2028

Team set up in 2026, pilots in early 2027, expands outlet coverage from 2H-2027 onwards; then the model keeps improving.

Financial gain and cost each year



	2026	2027	2028	2029	2030
Gain	–	+S$0.6m	+S$1.4m	+S$1.7m	+S$2.0m
Cost	−S$78k	−S$0.6m	−S$0.4m	−S$0.4m	−S$0.4m
Cumulative	−S$78k	−S$62k	+S$1.0m	+S$2.3m	+S$3.9m

REVENUE IMPACT

+S$5.0m extra revenue over five years

About 305,000 more cups a year by 2030

P1	Fewer walk-aways at peak	S$2.1m
P2	Staff and machines ready for the peak	S$1.1m
P3	Better-placed new stores	S$1.8m

COST IMPACT

S$3.0m saved

Less idle staff time, fresh-stock waste and machines

S$1.8m invested

S$0.6m one-off, then S$0.4m a year

FINANCIAL RESULT

S$3.9m net over five years

Breakeven **early 2028** · **S$1.6m** a year at full scale

Full financial model →

BEYOND THE NUMBERS — Brand value · A better customer experience · Calmer, better-looking stores — *Harder to measure, just as important*

SPEAKER NOTES · FINANCIAL IMPACT

What the three proposals are worth together, over five years, on top of today's business. All figures are our illustrative assumptions, not Chagee data.

Read the chart from left to right. In 2026 we only set up the project team, about S$78k. In 2027 we build the platform and pilot in the first half, so cost peaks at about S$0.6m; from the second half every outlet is covered. From 2028 the shared model keeps learning, so the green bars grow while running costs stay at about S$0.4m a year. The black line is the cumulative result: it bottoms out at −S$0.1m and turns positive in 2028.

Revenue: more demand captured and served. Proposal one recovers customers who would have walked away at peak, proposal two has staff and machines ready for the busy days, and proposal three places new stores where they sell more. Together about 305,000 more cups a year by 2030, S$5.0m of revenue over five years, counted in the benefit at gross margin.

Cost: proposal two saves S$3.0m in idle staff hours, wasted fresh stock and machines, against S$1.8m of investment.

Result: S$3.9m net over five years, breaking even in early 2028, and about S$1.6m a year once everything runs at full scale. The full model and its assumptions are in the appendix and the Excel workbook. About one minute.

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



SPEAKER NOTES · THANK YOU

Recap the chain and its owners: the CRM manager makes membership worth joining, the operations manager serves the demand reliably, the expansion manager opens where demand remains underserved. Each part rests on its own assumptions; no simulated uplift is carried from one to the next.

Likely questions: is the brigade always better? No; with spare staff fixed stations are as good. Is the forecast real? No, it is an illustration of the chain, built on stated assumptions.

APPENDIX

SUPPORTING DETAILS FOR THE PRESENTATION

FINANCIALS & SUPPORTING ILLUSTRATIONS FOR PROPOSALS

SPEAKER NOTES · APPENDIX

The appendix, for questions: the detail behind the main slides, in their order. The contents follow on the next slide.

The five-year model behind the financial impact

S$ THOUSANDS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
REVENUE IMPACT						
Proposal 1 · cups recovered from walk-aways (thousands)	–	42	91	107	118	357
Proposal 2 · extra cups served at the peak (thousands)	–	20	50	59	66	195
Proposal 3 · extra cups at better-chosen sites (thousands)	–	25	64	92	122	302
Additional revenue	**–**	**513**	**1,207**	**1,519**	**1,799**	**5,038**
Less ingredients and packaging	–	(236)	(555)	(699)	(828)	(2,317)
Gross profit from additional sales	**–**	**277**	**652**	**820**	**972**	**2,720**
COST IMPACT · LEANER OPERATIONS (PROPOSAL 2)						
Staff overhead saved	–	130	328	386	427	1,272
Fresh stock waste avoided	–	141	356	419	464	1,380
Tea machines avoided	–	37	91	103	111	342
Operating cost savings	**–**	**307**	**775**	**908**	**1,002**	**2,993**
Total benefit	**–**	**584**	**1,427**	**1,729**	**1,974**	**5,714**
ADDITIONAL COST FROM INITIATIVES						
Project team	78	312	–	–	–	390
Platform build	–	100	–	–	–	100
Data pipelines and cloud set-up	–	60	–	–	–	60
Security and PDPA review	–	20	–	–	–	20
Run team	–	–	240	240	240	720
Software maintenance	–	–	20	20	20	60
Cloud, licences and AI services	–	52	70	70	70	262
Switch perks (Proposal 1)	–	23	50	59	66	198
Total cost	**78**	**568**	**380**	**389**	**396**	**1,811**
Net financial impact	**(78)**	**16**	**1,047**	**1,340**	**1,578**	**3,903**
Cumulative net impact	*(78)*	*(62)*	*985*	*2,324*	*3,903*	

Key assumptions

Some based on Chagee's real online data; the rest are illustrative assumptions

BASELINE

44 outlets × 240,000 cups a year at S$5.90, growing 3% a year, plus the new outlets: the base the percentages below apply to. Gross margin 54%; wages rise 3% a year

ROLL-OUT

Pilots in the first half of 2027; every outlet from the second half. The model reaches its full effect by 2030

PROPOSAL 1

3% of cups lost at peak, 30% of them recovered; 2% of orders redirected, 50% of them with a S$0.50 perk

PROPOSAL 2

10 idle staff hours a week saved per outlet; fresh-stock waste of 3% of sales cut by 20%; 0.5% more cups at the peak

PROPOSAL 3

Two new outlets in 2027, then one a year, each selling 10% more than a conventional site pick

COSTS

Platform build S$100k, data set-up S$60k and security review S$20k in 2027; then a run team S$240k, cloud S$70k and maintenance at 20% of the build a year. Chagee's managers and store teams are existing staff: no extra cost

SPEAKER NOTES · THE FIVE-YEAR MODEL BEHIND THE FINANCIAL IMPACT

The five-year model behind the financial impact, in thousands of Singapore dollars. Revenue comes from extra cups: cups recovered from walk-aways with proposal one, extra cups served at the peak with proposal two, and extra cups at better-chosen sites with proposal three. Less ingredients and packaging, that gives the gross profit from additional sales.

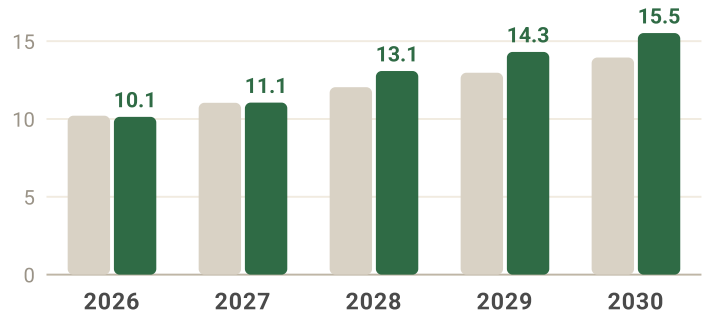
Proposal two also lowers operating costs: staff overhead, fresh-stock waste and tea machines avoided. Against that sit the initiatives' costs: the project team, the platform build, data and cloud set-up, a security and PDPA review, then a run team, maintenance, cloud and AI services, and the switch perks. The net impact turns positive in the first full year of roll-out and the cumulative impact by the year after. The key assumptions are on the right; they are illustrative and to be confirmed with Chagee.

With the proposals, Chagee Singapore profit grows to S$15.5m by 2030 with a 19.6% margin

The whole Singapore business, growing 3% a year with new outlets (49 by 2030), with and without the proposals.

S$ MILLIONS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
Outlets at year end	44	46	47	48	49	
Cups sold (millions)	10.6	11.2	12.0	12.7	13.4	59.9
REVENUE						
Baseline sales	62.3	65.6	69.9	73.5	77.3	348.6
Additional sales from the proposals	–	0.5	1.2	1.5	1.8	5.0
Total revenue	**62.3**	**66.1**	**71.1**	**75.0**	**79.1**	**353.6**
Cost of goods sold	(28.7)	(30.3)	(32.3)	(34.1)	(35.9)	(161.3)
Gross profit	**33.6**	**35.9**	**38.7**	**40.9**	**43.2**	**192.3**
Gross margin	*54.0%*	*54.2%*	*54.5%*	*54.6%*	*54.6%*	
OPERATING EXPENSES						
Staff	(8.6)	(8.9)	(9.3)	(9.7)	(10.2)	(46.8)
Rent and utilities	(9.2)	(9.4)	(9.8)	(10.0)	(10.2)	(48.6)
Delivery-app commissions	(5.6)	(5.9)	(6.3)	(6.6)	(7.0)	(31.4)
Equipment: tea machines avoided	–	0.0	0.1	0.1	0.1	0.3
AI platform and team	(0.1)	(0.6)	(0.4)	(0.4)	(0.4)	(1.8)
Total operating expenses	**(23.5)**	**(24.8)**	**(25.6)**	**(26.6)**	**(27.7)**	**(128.2)**
Outlet operating profit	**10.1**	**11.1**	**13.1**	**14.3**	**15.5**	**64.1**
Operating margin	*16.3%*	*16.7%*	*18.4%*	*19.1%*	*19.6%*	

FOR COMPARISON · WITHOUT THE PROPOSALS						
Outlet operating profit without the proposals	10.2	11.0	12.0	13.0	13.9	60.2
Operating margin without the proposals	*16.4%*	*16.8%*	*17.2%*	*17.6%*	*18.0%*	
Uplift from the proposals	**(0.1)**	**0.0**	**1.0**	**1.3**	**1.6**	**3.9**



REVENUE IN 2030

S$79.1m

up 27% on 2026, with 49 outlets

MARGIN IN 2030

19.6%

against 18.0% without the proposals

SPEAKER NOTES · FINANCIAL IMPACT: CHAGEE SINGAPORE FORECAST

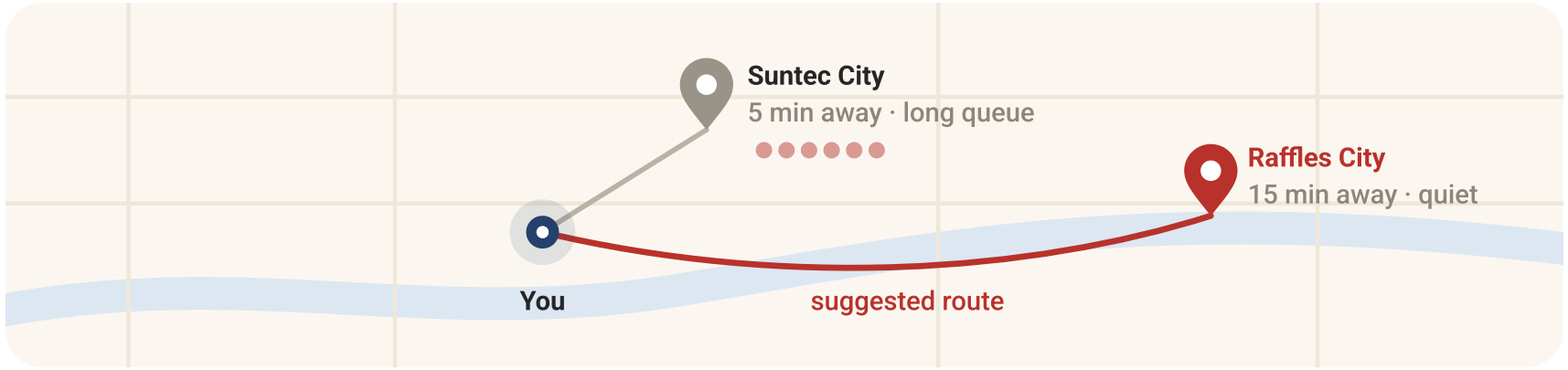
This is the whole Singapore business, not just the impact. We assume 3% organic growth in cups a year, wages rising 3% a year, two new outlets in 2027 and one a year after that, so revenue grows from S$62.3m in 2026 to S$79.1m in 2030 across 49 outlets.

Gross margin is 54%, in line with Chagee's reported corporate margin. Outlet costs are staff, rent and utilities, and delivery-app commissions. This is outlet-level profit, before head-office costs and the set-up cost of new outlets.

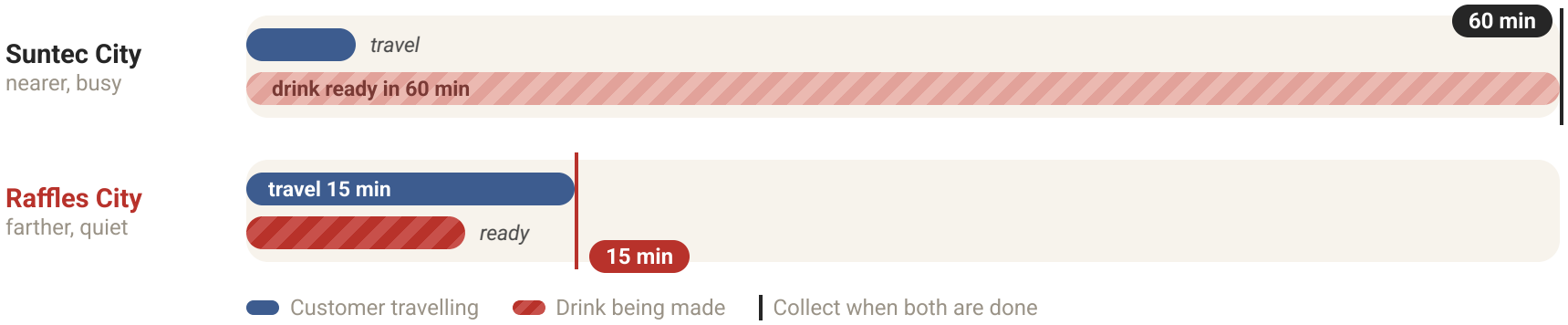
The green bars add the three proposals: extra sales, Proposal 2's savings, minus the cost of the initiatives. The uplift row matches the previous slide's net impact. By 2030, outlet profit is S$15.5m against S$13.9m without the proposals, and the margin rises from 18.0% to 19.6%. About forty-five seconds.

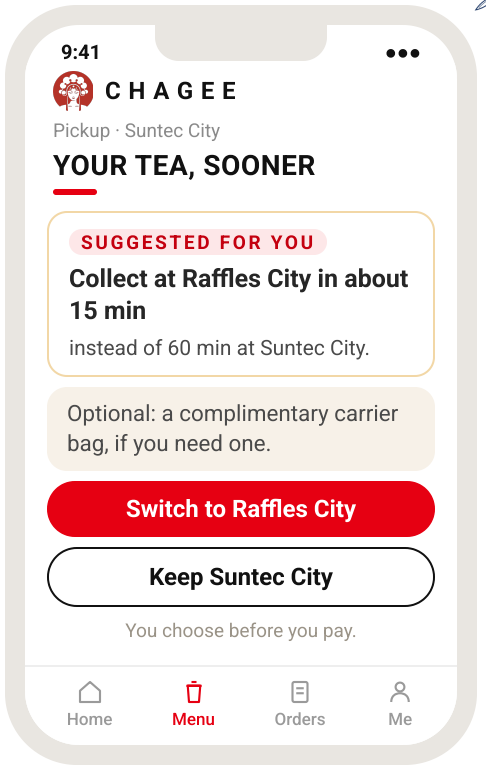
Your tea, sooner: AI routes pickups to a faster outlet

When one outlet is swamped, the app offers a nearby outlet that can hand over the drink sooner, and the customer chooses.



THE RACE TO YOUR DRINK





GENERATIVE AI MESSAGE · MOCK-UP

45 minutes sooner

Travel and preparation happen at the same time, so the customer collects as soon as both are done.

SPEAKER NOTES · THE PICKUP RACE (EARLIER VERSION)

Proposal one tackles online retailing: using data for better sales, loyalty and cost management. The idea is smart pickup routing.

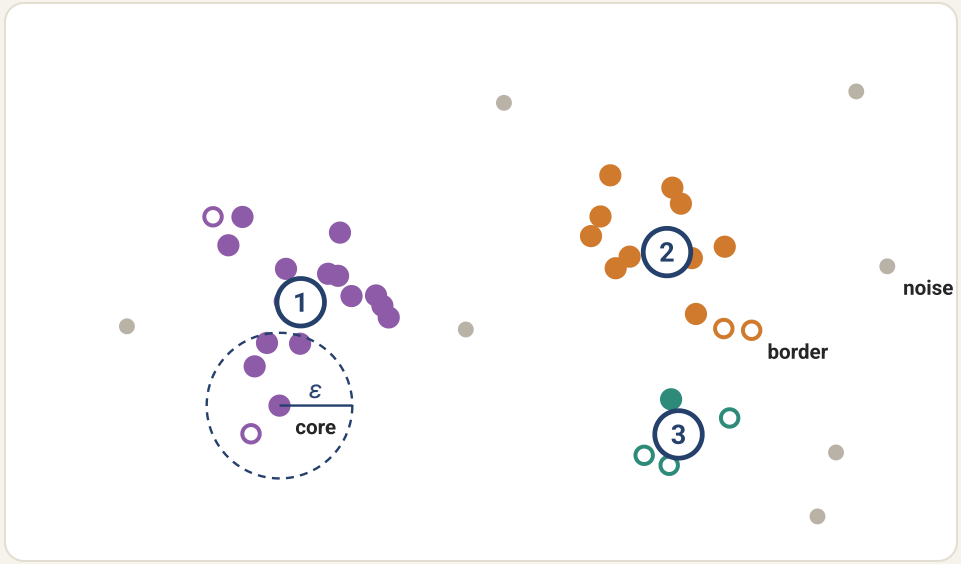
Picture a customer at peak. Suntec City is the nearest outlet, 5 minutes away, but it is swamped: the drink will not be ready for 60 minutes. Raffles City is farther, 15 minutes away, but quiet: the drink is ready in 10.

Watch the race. Travel and preparation happen at the same time, so you collect when both are done: 60 minutes at Suntec, 15 at Raffles. That is 45 minutes sooner for the customer, and the busy outlet's queue gets relief.

On the right is how it would look in the Chagee app: generative AI writes a short, clear message, with an optional complimentary carrier bag, and the customer chooses before paying. Nothing switches without consent. All times here are illustrative, not live Chagee data. About fifty seconds.

DBSCAN finds where searches bunch up; a demand model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 08 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

SPEAKER NOTES · HOW THE SITE FORECAST WORKS

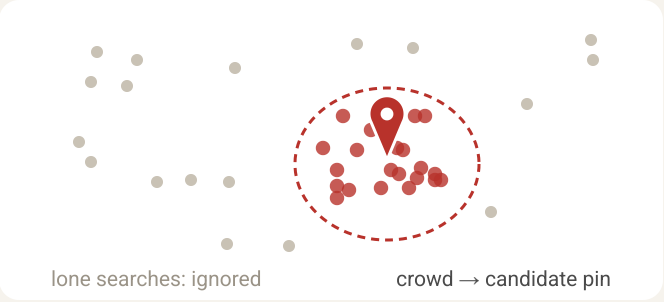
Backup for questions on slide 08. Left: DBSCAN in four moves on a toy set. A search is a core point if enough searches sit within the radius; clusters grow from core to core; border points join; the rest is noise. We do not choose the number of clusters.

Right: every simulated number, in the order the model uses it. The forecast is only as good as these assumptions; with Chagee's real app logs, the first two rows would be replaced by data and the rest would run as is.

DBSCAN picks the spot; a demand model counts the cups

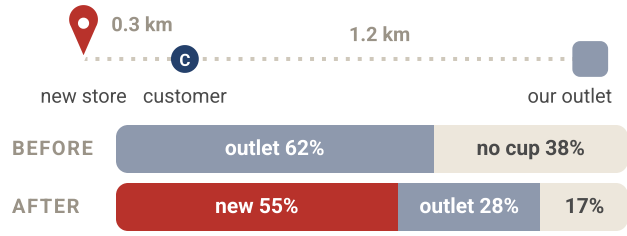
Two separate jobs. DBSCAN never predicts sales: it only finds where unserved searches crowd together.

1 Find the crowd — DBSCAN



Keep the searches with no outlet within 1 km. Ask each one: **are at least 15 searches within 0.6 km of me?** Crowds that touch merge into one; lone searches are ignored. The centre of each crowd is a candidate site.

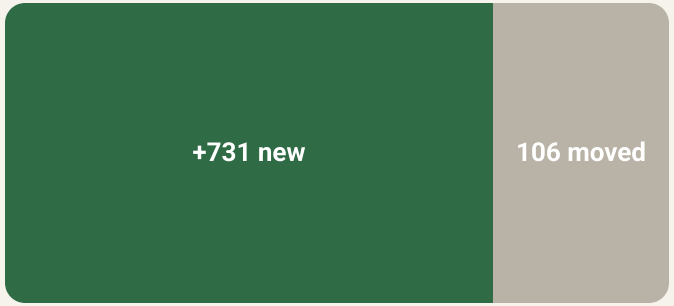
2 Feel the pull — DEMAND MODEL



The closer a store, the more of a customer's cups it gets. Some cups are never bought at all.

When the new store opens, it gets 55% of this customer's cups: **34% taken from our outlet** + **21% never bought before**.

3 Count new, not moved



Add up every customer's share: a new store at the top site sells **837 cups a day**. 106 of them would otherwise have gone to our existing outlets, so **+731 are new to Chagee**.

Strong site if at least 60% of its cups are new; weak below 40%.

IN ONE LINE DBSCAN says **where** to look; the demand model says **how many** cups a store there would sell, and how many of those are really new.

SPEAKER NOTES · THE SITE FORECAST, SIMPLY

The intuition, in three moves. One: DBSCAN looks only at searches from people with no outlet within walking distance, and asks each one whether enough other searches sit close by. Where they crowd together, the centre of the crowd becomes a candidate site; lone searches are ignored, and nobody chooses the number of crowds.

Two: DBSCAN does not predict sales. The cups come from a separate demand model. Every store pulls on a customer, nearer pulls harder, and not buying at all pulls too; the customer's cups split in proportion to those pulls. Take one customer, 0.3 km from the new store and 1.2 km from our outlet. Before the store opens, 62% of their potential cups go to the outlet and 38% are never bought. After it opens, the new store gets 55%: 34 points it takes from our outlet, which are moved cups, and 21 points that would not have been bought at all, which are new cups.

Three: add those shares up. At the top site the new store sells about 837 cups a day, but 106 of them would have gone to our own outlets anyway, so about 731 are new to Chagee. That share of new cups decides strong, mixed or weak. The model is calibrated so the average outlet sells 556 cups a day, and the searches are synthetic: with Chagee's real app logs, the method runs unchanged.